

Algebra & Polynomials

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions with method lines. Curriculum: MTH1W (Grade 9 de-streamed math).

1. [L1 · Like terms] $7x + 2x - 3x = 6x$.
2. [L1 · Substitution] $5(3) - 4 = 15 - 4 = 11$.
3. [L1 · Add polynomials] $(3a + 4a) + (2 - 6) = 7a - 4$.
4. [L1 · Multiply by constant] $6 \cdot 2x - 6 \cdot 3 = 12x - 18$.
5. [L1 · Divide by constant] $20x/4 + 8/4 = 5x + 2$.
6. [L1 · Exponent law] Add exponents: $x^{6+3} = x^9$.
7. [L2 · Like terms] $(5x - 2x) + (-3y + 8y) - 4 = 3x + 5y - 4$.
8. [L2 · Substitution] $3(4) - 2(5) = 12 - 10 = 2$.
9. [L2 · Subtract polynomials] $6x^2 - 2x^2 = 4x^2$; $2x - (-3x) = 5x$; $-5 - 1 = -6$.
 $= 4x^2 + 5x - 6$.
10. [L2 · Multiply by monomial] $4x \cdot 3x - 4x \cdot 2 = 12x^2 - 8x$.
11. [L2 · Divide by monomial] $18x^2/6x - 12x/6x = 3x - 2$.
12. [L2 · Exponent laws] Coefficients $3 \times 5 = 15$; add exponents: $15x^6$.
13. [L3 · Find the error] Adding like terms adds the coefficients only; the variable's power is unchanged.
 $2x + 3x = 5x$, not $5x^2$.
14. [L3 · Area (substitute)] $A = (2x)^2 = 2^2x^2 = 4x^2$.
15. [L3 · Perimeter (polynomial)] $P = 2[(3x + 2) + (x - 1)] = 2(4x + 1) = 8x + 2$.
16. [L3 · Expand & simplify] $6x - 3 - 2x + 8 = 4x + 5$.
17. [L3 · Divide polynomial] Divide each term by $5x$: $2x^2 - 3x + 1$.
18. [L3 · Power of a monomial] Square each factor: $2^2a^6b^4 = 4a^6b^4$.